JOHN ASARO

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Saratoga Springs, NY 10007 (open to relocation)

EDUCATION

CONNECTICUT COLLEGE, New London, CT — May 2026 | GPA: 3.83
Majors: Computer Science, Psychology
Honors/Awards: Dean's High Honors, Junior/Senior Computer Science Award, ECTA 2025 Best Poster Award

SELECTED WORK EXPERIENCE

AI Researcher, Autonomous Agent Learning Lab — October 2024 - May 2026
New London, CT

- Developed evolutionary learning systems for autonomous agents in simulated robotics and game environments, coauthoring two peer-reviewed papers accepted to ECTA 2025, one of which received the Best Poster award
- Collaborated with lab members to prototype and iterate on AI experiments, contributing code to support ongoing autonomous agent research

Computational Social Choice Researcher, Summer Science Research Institute (SSRI) — May 2024 - July 2024
New London, CT

- Conducted applied research on algorithmic decision making systems, focusing on how to aggregate individual preferences to maximize overall social benefit
- Designed and implemented a novel aggregation algorithm in Python
- Demonstrated that the proposed algorithm consistently outperformed existing approaches in measured social benefit across both single-selection and multi-selection scenarios

Software Developer, Zummo Bike — August 2023 - December 2023
Remote

- Developed backend components for a data entry software used by a nonprofit, improving the reliability and structure of inventory and client data management
- Built a REST API with Flask to support CRUD workflows and power a Svelte-based frontend for real-time data entry and updates
- Integrated Stripe and Airtable APIs to enable payment processing and automated data synchronization, reducing manual operational overhead

PROJECTS

Viz Ikemen — July 2025 - October 2025

- Built Viz-Ikemen, a reinforcement learning environment for training vision-based AI agents in a real-time game environment
- Verified end-to-end environment functionality by training a Deep Q-Network agent with sparse rewards, achieving a 100% win rate against a baseline opponent once observation and action interfaces were working perfectly
- Integrated the game engine with an external RL pipeline by adding real-time visual streaming and action control, enabling scalable, headless training across multiple parallel instances

PUBLICATIONS

Decentralized Evolution of Hexapod Gaits with Independent Leg Controllers
Gary B Parker, John Asaro, Jim O'Connor
IJCCI Conference on Evolutionary Computation and Theory and Applications (ECTA), 2025

Niching Agents in The Core
Gary B Parker, Jim O'Connor, John Asaro
IJCCI Conference on Evolutionary Computation and Theory and Applications (ECTA), 2025

SKILLS

Languages: Python, C++, C#, Java, Javascript, Go, Lua, and Scheme
Systems/Tools: Linux, Git, SQL, REST APIs, Azure, Flask, PyTorch, Gymnasium, Matplotlib, NumPy